Solutions

A14-N01

$252=1\cdot 198+54$; $198=3\cdot 54+36$; $54=1\cdot 36+18$; $36=2\cdot 18+0$, so gcd=18.

A14-N02

$(r,s,t)=(252,1,0),(198,0,1),(54,1,-1),(36,-3,4),(18,4,-5)$, so $18=4\cdot 252-5\cdot 198$.

A14-N03

Divide both sides and modulus by gcd(14,100)=2, obtaining $7x\equiv 15 \pmod{50}$. Since $50=7\cdot 7+1$, $1=50-7\cdot 7$, so $7^{-1}\equiv -7\equiv 43 \pmod{50}$. Then $x\equiv 15\cdot 43=645\equiv 45 \pmod{50}$. The distinct solutions modulo 100 are 45 and 95; direct substitution verifies both.

A14-N04

gcd(6,9)=3 does not divide 5, so no solution exists.

A14-N05

Put $x=2+6t$. Then $6t\equiv 3 \pmod 9$, so $t\equiv 2 \pmod 3$ and $x\equiv 14 \pmod{18}$.

A14-N06

gcd(6,9)=3 does not divide $4-2=2$, so no solution exists.

A14-N07

$13=(1101)_2$. Start r=1; for each bit, square r and multiply by 7 if the bit is 1, reducing modulo 23: bit 1 gives 7; bit 1 gives $7^2\cdot 7\equiv 21$; bit 0 gives $21^2\equiv 4$; bit 1 gives $4^2\cdot 7\equiv 20$. For exponent 0 no bits are processed and the result is 1 mod 23.

A14-N08

$15-1=2\cdot 7$ and $2^7 \bmod 15=8$ is neither 1 nor 14. No squares remain, so 2 is a witness.

A14-N09

$\phi(143)=(11-1)(13-1)=120$. Euclid gives $120=17\cdot 7+1$, hence $1=120-17\cdot 7$ and the inverse of 7 modulo 120 is $-17\equiv 103$. Thus $d=103$; verify $7\cdot 103=721\equiv 1 \pmod{120}$.

A14-N10

For encryption $7=(111)_2$, the left-to-right accumulator is 9,14,48, so $9^7\equiv 48 \pmod{143}$. For decryption $103=(1100111)_2$: start at 1, square at each bit, multiply by 48 on a 1, and reduce modulo 143. The successive accumulators are 48,53,92,27,100,92,9, yielding $48^{103}\equiv 9 \pmod{143}$.

A14-X01

$55=1\cdot 34+21$; $34=1\cdot 21+13$; $21=1\cdot 13+8$; $13=1\cdot 8+5$; $8=1\cdot 5+3$; $5=1\cdot 3+2$; $3=1\cdot 2+1$; $2=2\cdot 1+0$. Quotient 1 makes each preceding remainder the sum of the next two, so consecutive Fibonacci numbers attain logarithmically many divisions. Each long division still has a bit-dependent cost.

A14-X02

$M=4\cdot 5\cdot 7=140$, $M_i=(35,28,20)$, with respective inverses (3,2,6). Thus $x\equiv 1\cdot 35\cdot 3+2\cdot 28\cdot 2+3\cdot 20\cdot 6=577\equiv 17 \pmod{140}$. The remainders of 17 modulo 4,5,7 are 1,2,3.

A14-X03

$d=\gcd(4,6)=2$ divides $3-1=2$. Put $x=1+4t$; then $4t\equiv 2 \pmod 6$, so $2t\equiv 1 \pmod 3$, $t\equiv 2 \pmod 3$, and $x\equiv 9 \pmod{12}$. In the second system 2 does not divide $2-1=1$, so no solution exists.

A14-X04

The powers of 2 modulo 11 are 2,4,8,5,10,9,7,3,6,1. The first return to 1 is at exponent $10=\phi(11)$, so 2 is primitive and $x\equiv 7 \pmod{10}$.

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A14-X05

$9-1=2^3 \cdot 1$. Among the units, the strong test accepts only a=1 and a=8, immediately equal to 1 and −1 respectively. The other units 2,4,5,7 fail, and 3,6 are non-units. There are 2 liars, exactly (9−1)/4.

A14-X06

slow=f(2)=5 and fast=f(f(2))=f(5)=26. The gcd is gcd(|5−26|,91)=gcd(21,91)=7, yielding $91=7 \cdot 13$. If gcd=91, this round failed to separate a factor; retry with a different seed or function rather than reporting 91 as a nontrivial factor.

A14-X07

Floyd’ s (slow,fast,d) values are (5,26,1), (26,7474,1), (677,871,97); round 3 returns 97. Checkpoint comparisons (y,x,d) are (5,26,1), (26,677,1), (26,7474,1), (7474,2839,1), (7474,871,1), (7474,1848,97); the last returns 97. For $0 \le k \le 97$, $P(\text{no collision}) = \prod_{j=0}^{k-1}(1-j/97) \le \exp(-k(k-1)/194)$, and the probability is zero for k>97. A fixed polynomial sequence need not follow the independent-uniform model, so this is not a direct guarantee for its success probability.

A14-X08

U(12)={1,5,7,11}, and every element squares to 1 modulo 12. There is no element of order 4, so the group is not cyclic. Powers of 2 modulo 9 are 2,4,8,7,5,1; its order is $6=\phi(9)$, so 2 is primitive.

A14-X09

$4^2=16 \equiv 1 \pmod{15}$ and 4 is neither 1 nor 14 modulo 15. gcd(4−1,15)=3 and gcd(4+1,15)=5, giving $15=3 \cdot 5$.

A14-X10

$\phi(55)=4 \cdot 10=40$, and $3 \cdot 27=81 \equiv 1 \pmod{40}$. The message has gcd(5,55)=5, and $c=5^3 \equiv 15 \pmod{55}$. The squares are $15^2 \equiv 5$, $15^4 \equiv 25$, $15^8 \equiv 20$, $15^{16} \equiv 15$. Since 27=16+8+2+1, $15 \cdot 20 \equiv 25$, $25 \cdot 5 \equiv 15$, and $15 \cdot 15 \equiv 5$. Modulo 5 the component remains 0; modulo 11 it is nonzero and Fermat’ s theorem with $3 \cdot 27 \equiv 1 \pmod{10}$ restores it. CRT reconstructs 5.

A14-X11

Since gcd(2,55)=1 and $2^3=8$, $c' \equiv 15 \cdot 8 \equiv 10 \pmod{55}$, and $10^{27} \equiv 10 \equiv 2 \cdot 5 \pmod{55}$. A third party can transform the ciphertext into one encrypting twice the original message. This demonstrates multiplicative malleability, not universal plaintext disclosure in every protocol.

A14-X12

The distinct cosets are {0,4,8}, {1,5,9}, {2,6,10}, and {3,7,11}. They are disjoint and cover Z_{12}, giving $12=4 \cdot 3$. Also $72=8 \cdot 9$, $\phi(8)=8-4=4$ and $\phi(9)=9-3=6$, so $\phi(72)=24$.

A14-X13

Since $25=5^2$ for odd prime 5, the only roots are ± 1, namely 1 and 24. Modulo $15=3 \cdot 5$, the two CRT components can independently choose signs, giving four roots 1,4,11,14.

A14-X14

$12^3=1728 \equiv 23 \pmod{55}$, so m=23 and $(m,\sigma)=(23,12)$ satisfies $\sigma^3 \equiv m$. The method chooses σ before deriving m; it does not prove the ability to sign an arbitrary message fixed in advance.

A14-X15

$341=11 \cdot 31$ and $2^{10}=1024 \equiv 1 \pmod{341}$, so $2^{340} \equiv 1$ and Fermat accepts. Write $340=2^2 \cdot 85$: $2^{85} \equiv 32$ and $32^2 \equiv 1$. The sequence 32,1 starts at neither 1 nor −1 and never reaches −1, so Miller–Rabin rejects.

A14-X16

$561=3 \cdot 11 \cdot 17$ and 560 is divisible by 2,10,16. For gcd(a,561)=1, Fermat's theorem yields $a^{560} \equiv 1$ in every prime component; CRT gives the congruence modulo 561. For base 2 with $560=2^4 \cdot 35$, the strong sequence is 263,166,67,1. It does not start at 1 and never reaches 560, so the test rejects.

A14-X17

2047=23 · 89; 2046=2 · 1023, $2^{11} \equiv 1$, and 1023=11 · 93, so $2^{1023} \equiv 1$ and that round passes. The randomized 1/4 error bound does not apply to always choosing base 2; it requires uniform sampling from the specified range and independent rounds for repeated trials.

A14-X18

Invariant: $r \cdot a^b \equiv 7^{19} \pmod{23}$. For b=2q the updated product is $r(a^2)^9$; for b=2q+1 it is $(ra)(a^2)^9$. Both equal the old product. Starting from (1,7,19), the states are (7,3,9), (21,9,4), (21,12,2), (21,6,1), (11,13,0); the result is 11. Exponent 0 returns 1 mod n, which is 0 for n=1. No inverse is used, so non-coprime bases are allowed.

A14-X19

g=gcd(24,60)=12. Divide to obtain $2x \equiv 3 \pmod 5$; inverse 3 gives $x_0=4$, step=5, count=12. The tuple (4,5,12) represents 4,9,…,59. For $0x \equiv 0 \pmod 9$ use (0,1,9), representing 0 through 8. A tuple takes O(log(n+1)) bits, while listing g terms needs Ω(g) output operations; computation cost is separate.

A14-X20

The order of 2 modulo 11 is 10. Because $2^{10} \equiv 56 \not\equiv 1 \pmod{121}$, its order modulo 121 is not 10. That order is a multiple of 10 and divides $\phi(121)=110$, leaving 110. It equals the size of U(121).

A14-X21

65=5 · 13, s=6,d=1. Initial ±1 yields 1 and 64. For $a^2 \equiv -1$ the roots are 2,3 modulo 5 and 5,8 modulo 13; CRT combines them into 8,18,47,57. The full set is {1,8,18,47,57,64}; only 4 of the 62 legal bases 2,…,63 are liars, so the probability is 2/31. Base 2 has checked chain 2,4,16,61,16,61 without 64 and is a witness.

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A14-X22

$4^{14}<10^{9}\le 4^{15}$, so 15 is the smallest choice certified by the universal bound 4^{-k}. This does not rule out lower actual error with fewer rounds on some inputs. For a fixed composite $\varepsilon=4^{-15}=1/1073741824$. With the stated prior and prime acceptance, Bayes gives at most $99\varepsilon/(1+99\varepsilon)=99/(1073741824+99)\approx 9.22\times 10^{-8}$. A fixed base 2 is not independently random: 2047 passes it every time, so repeated fixed tests do not multiply the error bound.

A14-X23

Choose k as the least multiple of λ at least max(μ,1), so $k\le \mu+\lambda$. Both x_k and x_{2k} lie on the cycle and their index difference k is divisible by λ, hence they meet. For n=15 the first round is (slow,fast,gcd)=(2,6,1), the second is (6,6,15). The value d=n is not a proper factor; change seed or function to retry, and report no factor found if the budget expires.

A14-X24

If $r=a \bmod b\le b/2$ the bound holds at once; otherwise $b/2<r<b$ and $b \bmod r=b-r<b/2$, so it holds within two divisions. 391=299+92, 299=3 · 92+23 and 92=4 · 23; back-substitution gives 23=4 · 299−3 · 391. Because gcd=23≠1, 299 has no inverse modulo 391.

References and attribution

Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein. Introduction to Algorithms, 3rd ed., Chapter 31. MIT 6.046J Fall 2004 Lecture 7, p. 33, and Fall 2005 Lecture 8, p. 13: inverses modulo a prime.

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